

Module 1

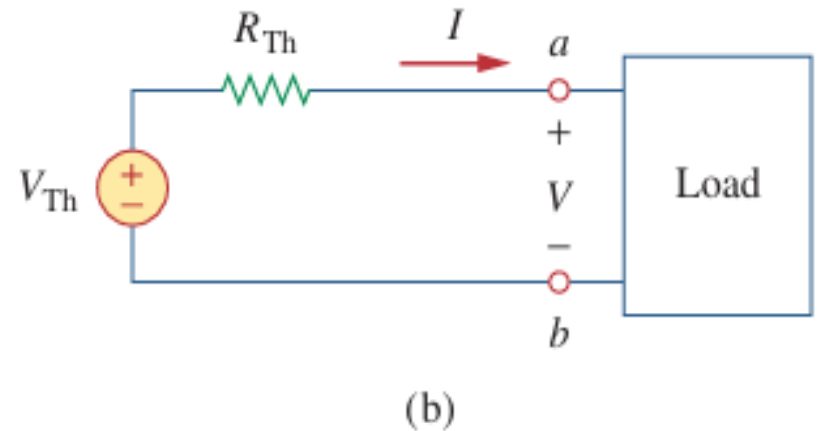
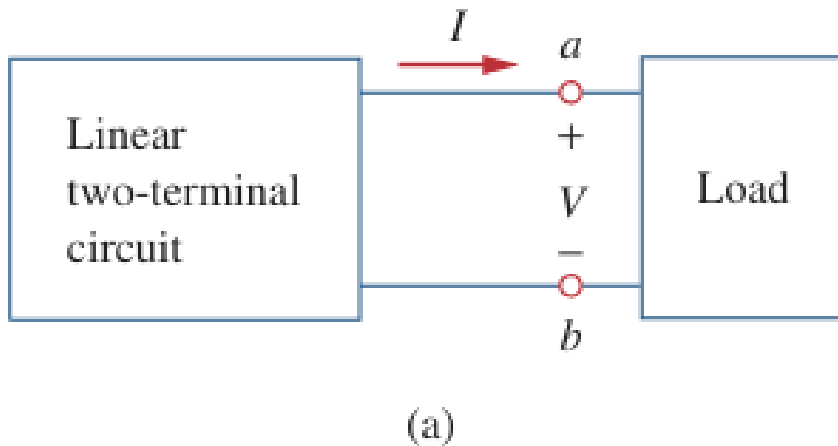
DC Circuit Analysis and Theorems

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Thevenin's Theorem

Thevenin's theorem states that a linear two-terminal circuit can be **replaced by an equivalent circuit consisting of a voltage source V_{th} in series with a resistor R_{th}** , where V_{th} is the open-circuit voltage at the terminals and R_{th} is the input or equivalent resistance at the terminals when the independent sources are turned off.

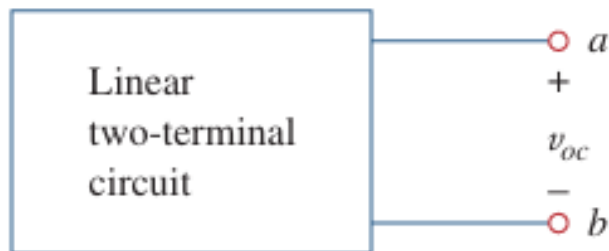


Two circuits are said to be equivalent if they have the same voltage-current relation at their terminals.

Thevenin's Theorem

Thevenin's equivalent voltage V_{th} and Thevenin's resistance R_{th} :

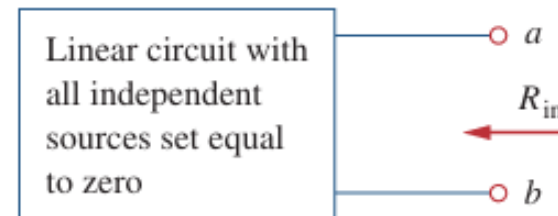
1. V_{th} is the open-circuit voltage across the terminals.
2. R_{th} is the input resistance at the terminals when the independent sources are turned off.



$$V_{Th} = v_{oc}$$

(a)

$$V_{Th} = v_{oc}$$



$$R_{Th} = R_{in}$$

(b)

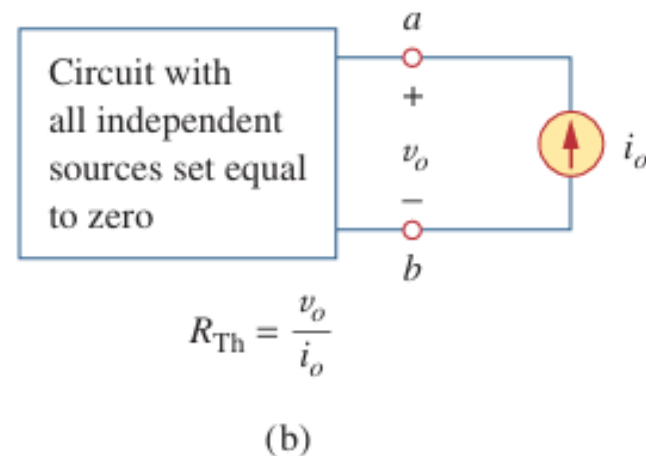
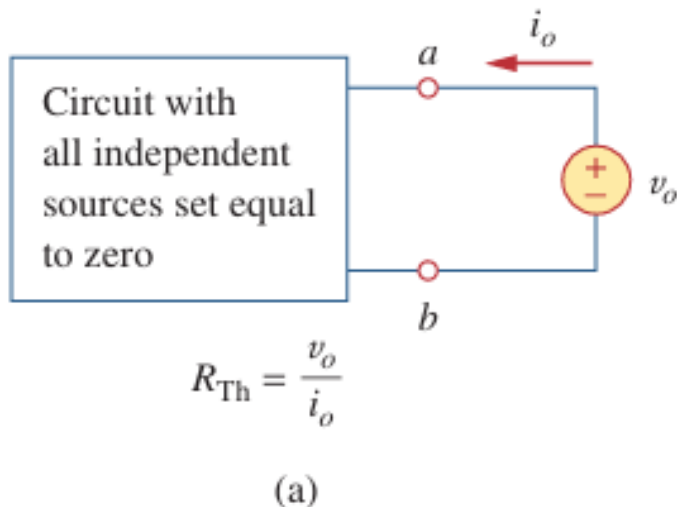
$$R_{Th} = R_{in}$$

Thevenin's Theorem

While finding the Thevenin's resistance , we need to consider two cases:

Case1: If the network has no dependent sources, we turn off all independent sources. R_{th} is the input resistance of the network looking between terminals a and b.

Case 2: If the network has dependent sources, we turn off all independent sources. Dependent sources are not to be turned off because they are controlled by circuit variables. We apply a voltage source at terminals a and b and determine the resulting current or insert a current source at terminals a and b and find the terminal voltage.

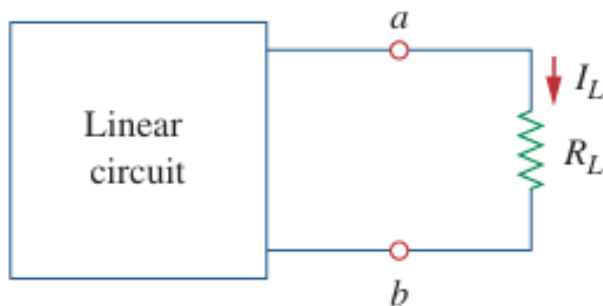


Thevenin's Theorem

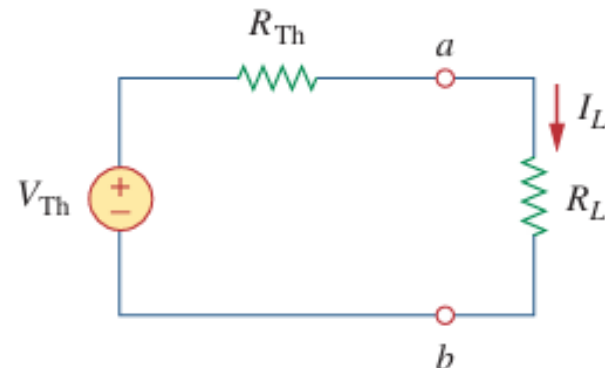
A large circuit may be replaced by a single independent voltage source and a single resistor.

A linear circuit with a variable load can be replaced by the Thevenin equivalent, exclusive of the load.

Consider a linear circuit terminated by a load R_L , the current through the load I_L and the voltage V_L across the load are easily determined once the Thevenin equivalent of the circuit at the load's terminals is obtained.



(a)



(b)

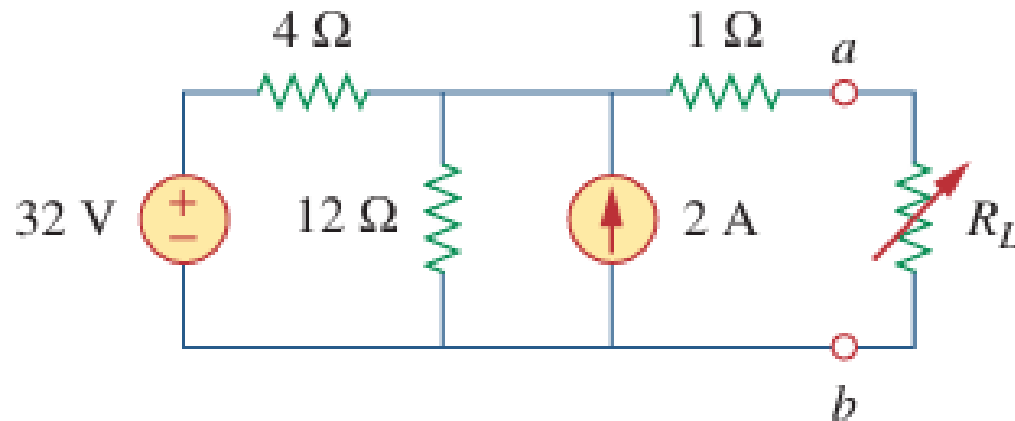
$$I_L = \frac{V_{Th}}{R_{Th} + R_L}$$

$$V_L = R_L I_L = \frac{R_L}{R_{Th} + R_L} V_{Th}$$

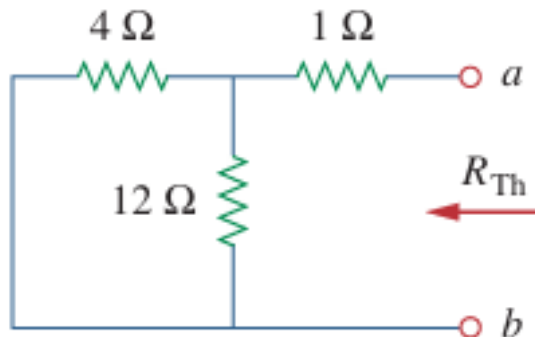
Thevenin's Theorem

Find the Thevenin's equivalent circuit of the circuit shown below, to the left of the terminals.

Then find the current through $R_L = 6\ \Omega$.



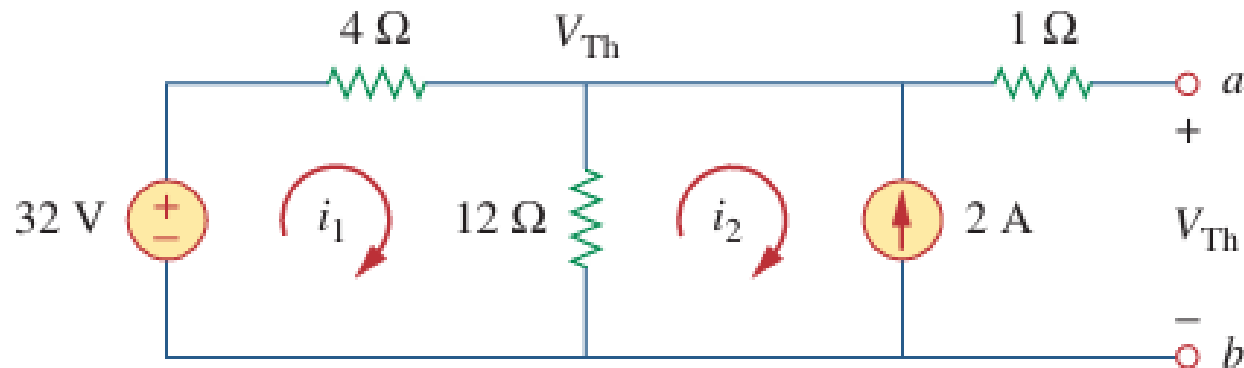
Find R_{th} by removing all the independent sources.



$$R_{Th} = 4 \parallel 12 + 1 = \frac{4 \times 12}{16} + 1 = 4\ \Omega$$

Thevenin's Theorem

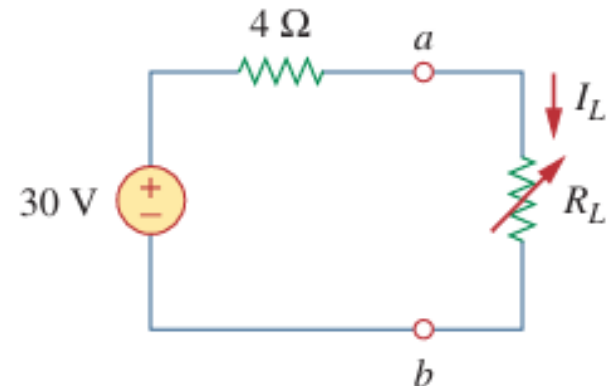
Find V_{th} by removing the load and finding voltage at the terminals a and b.



$$-32 + 4i_1 + 12(i_1 - i_2) = 0, \quad i_2 = -2 \text{ A}$$

$$i_1 = 0.5 \text{ A.}$$

$$V_{Th} = 12(i_1 - i_2) = 12(0.5 + 2.0) = 30 \text{ V}$$



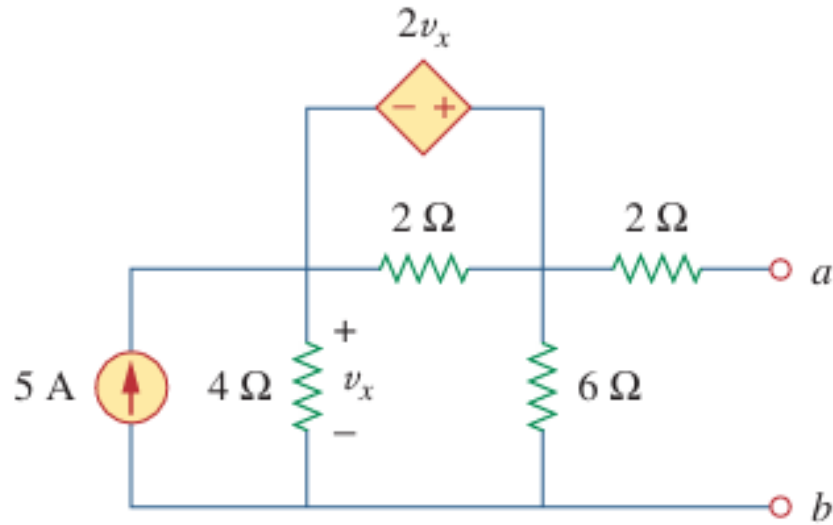
When $R_L = 6$,

$$I_L = \frac{30}{10} = 3 \text{ A}$$

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{30}{4 + R_L}$$

Thevenin's Theorem

Find the Thevenin's equivalent of the circuit at terminals a-b.



1. To find R_{th} we set the independent source equal to zero but leave the dependent source alone.
2. However, we excite the network with a voltage source v_o connected to the terminals.
3. For easy calculation set $v_o = 1$ V.
4. Let us now find the current i_o through the terminals, and then obtain $R_{th} = v_o/i_o$

Next class: Norton's Theorem